SOIL EROSION AND CONSERVATION · NOTE CM05

Grassed Waterways — Design, Velocities, Establishment and Maintenance

Course topics: SWCE4.5 L14, SWCE-501 T10

Role

Vegetated waterways are natural or constructed channels lined with erosion-resistant grass that carry surface water off cropland. They are the foundation of water management for most erosion-control practices (all except level terraces and contour bunds) — terrace systems, contour furrows, diversion channels and the emergency spillways of farm ponds. Establish the grass **before** water is turned into the waterway.

Design

- Carry the maximum runoff of a **10-year recurrence storm**, estimated by the rational method. On long waterways estimate the flow for several reaches, since the drainage area grows; design short channels for the outlet flow.
- **Shape:** trapezoidal, triangular or parabolic. When shaping natural depressions use a **parabolic** section (closest to natural channels). Triangular channels are easy to build with a V-ditcher and suit constructed channels. Broad-bottom trapezoids need less excavation than parabolic or triangular sections of the same capacity.
- **Location:** in natural drainageways (usually deep, fertile soil); without a natural depression, along field boundaries.

Design velocity	Grass condition
Low: 1 to 1.5 m/s	Sparse cover
Medium: 1.5 to 1.8 m/s	Good-quality cover
High: 2 to 2.5 m/s	Excellent sod only — use with extreme caution; avoid where possible

- Uniform grass cover is important for stability.
- **Grade:** set by the land slope; should not exceed 10% and is best kept within 5%. A diversion ditch across the slope intercepts runoff and leads it to a safe outlet.
- **Dimensions:** by Manning's formula with n ≈ 0.04 for grassed waterways; side slopes 4 : 1 or flatter; about 15 cm freeboard for sediment build-up and changes in roughness.

Establishing the grass

- **Grasses:** sod-forming, with deep, dense roots, adapted to the soil and site — one grass or a simple mixture of 2–3; salt- or alkali-tolerant where needed.
- **Preparation:** shape the channel to design width and depth; in natural drainageways fill small gullies and remove trees and shrubs; prepare a good seedbed and raise fertility by working manure in (nutrients and organic matter).
- **Planting:** solid or strip sod — can carry runoff at once if well anchored, but costly; or seeding — the commonest method, 1 to 2.5 cm deep with a drill, then packed with a roller.

Maintenance

- Take special care while the grass establishes: inspect after every rain, fill washouts with topsoil, and reseed or patch with sod.
- Do not use the waterway as a road; keep livestock out until the grass is well established (controlled grazing later, when the soil is not wet). Fertilise — especially with nitrogen — to keep growth vigorous.
- The commonest cause of waterway failure is **abuse and neglect**.

Sources: ICAR eCourse 'Soil and Water Conservation Engineering' — Dr A. Mishra, reviewed by Dr B. C. Mal (IIT Kharagpur); AgriMoon e-book. USDA-NRCS Conservation Practice Standard 412, Grassed Waterway. Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Grassed waterways are designed for the runoff of a storm with recurrence interval

(a) 2 yr (b) 5 yr
(c) 10 yr (d) 100 yr

Q2. For shaping natural depressions the preferred waterway section is

(a) triangular (b) rectangular
(c) parabolic (d) trapezoidal

Q3. Permissible velocity for a waterway with good-quality grass cover is

(a) 0.5–1 m/s (b) 1.5–1.8 m/s
(c) 2–2.5 m/s (d) 3–4 m/s

Q4. Grade of a grassed waterway should not exceed

(a) 1% (b) 5%
(c) 10% (d) 20%

Q5. Manning's n for grassed waterways is about

(a) 0.013 (b) 0.025
(c) 0.04 (d) 0.15

Q6. Freeboard in grassed waterways is about

(a) 5 cm (b) 15 cm
(c) 30 cm (d) 50 cm

Q7. The commonest cause of waterway failure is

(a) flood (b) drought
(c) abuse and neglect (d) pests

Answers

Q1 — (c) 10 yr
Q2 — (c) parabolic
Q3 — (b) 1.5–1.8 m/s
Q4 — (c) 10% — *and is best kept within 5%.*
Q5 — (c) 0.04
Q6 — (b) 15 cm
Q7 — (c) abuse and neglect